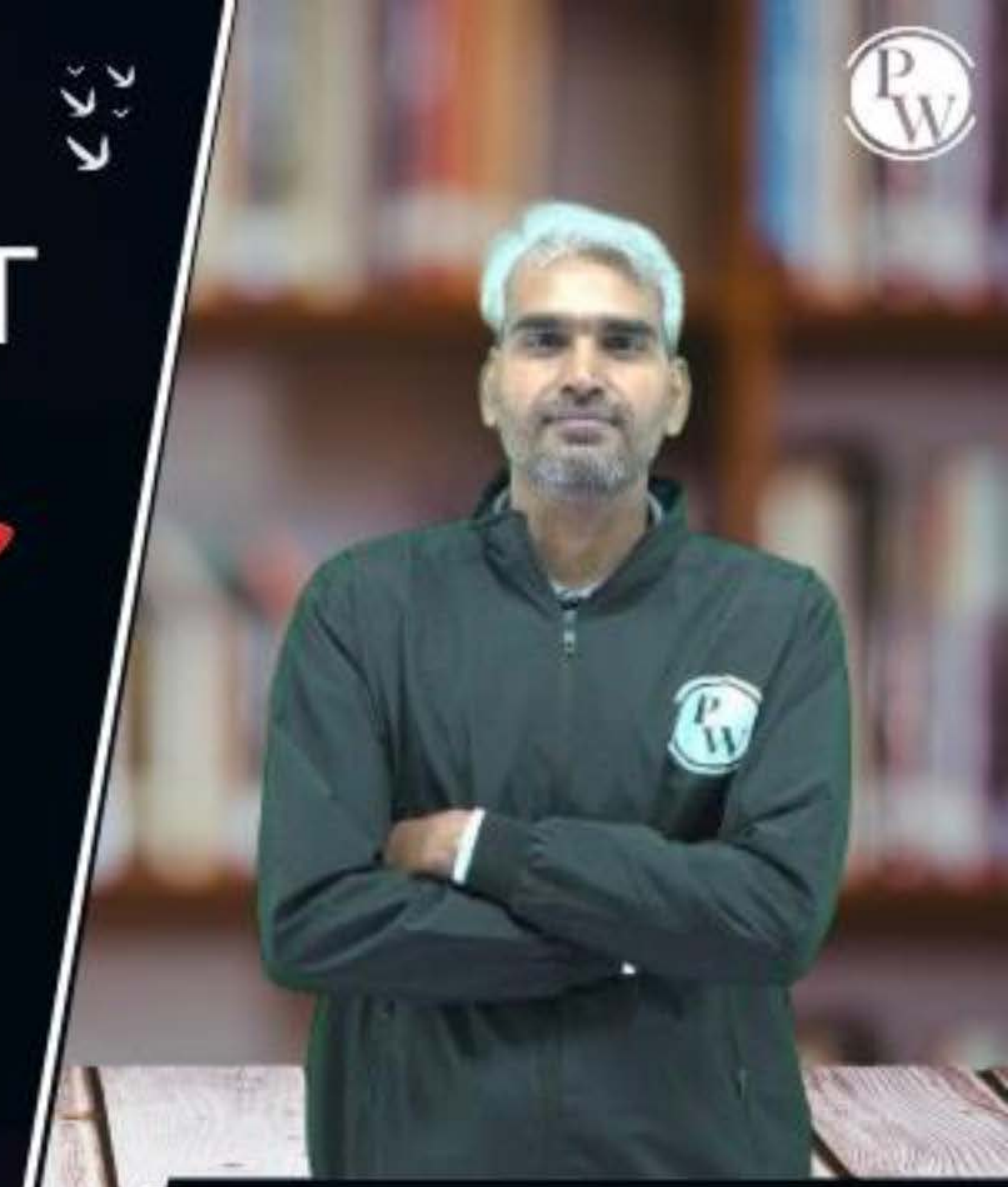
Computer Science & IT

Discrete Mathematics

Graph Theory

Lecture No. 05



By- Vishal Sir

Recap of Previous Lecture

Topic: Sum of degree theorem

Topic: Degree sequence

Topic: Graphic

Topic: Havel Hakimi's algorithm

Topic: Complement of a graph

Slide

Topics to be Covered

Topic — Graph isomorphism

Topic — Self-complementary graph

Topic — Planar graphs

Slide

Topic : Graph Isomorphism

Two graphs are isomorphic if and only if they have exact same properties {Shape & name need not be same}

- Two graphs G and G' are said to be isomorphic if there exists a function f: V(G)→V(G') such that

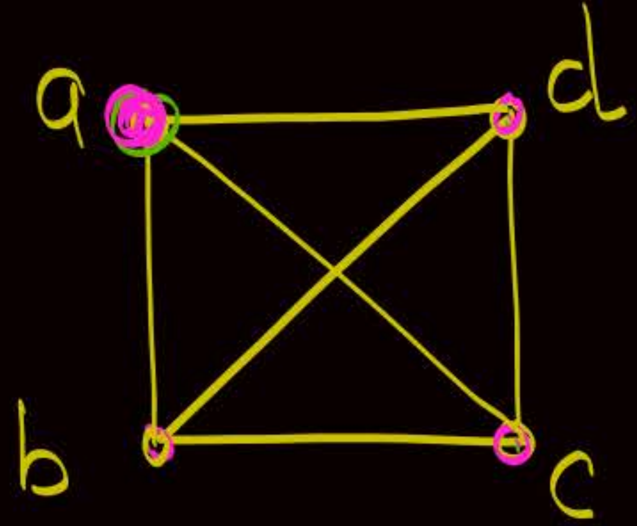
I. f is bijective i.e. $|V(G)| = |V(G')|$

II. f preserves adjacency of vertices

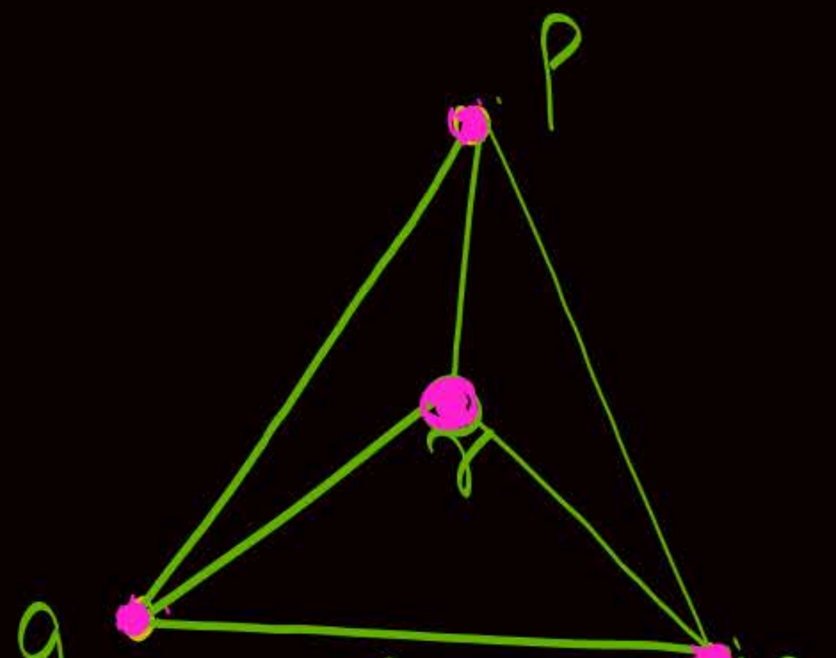
i.e. if two vertices $v_1, v_2 \in V(G)$ are adjacent to each other in graph G, then their images w.r.t. function f i.e. $f(v_1), f(v_2) \in V(G')$ must also be adjacent and vice-versa.

$\Rightarrow \therefore |E(G)| = |E(G')|$

Check whether G_1 & G_2 are isomorphic or not?



$G_1 (K_4)$

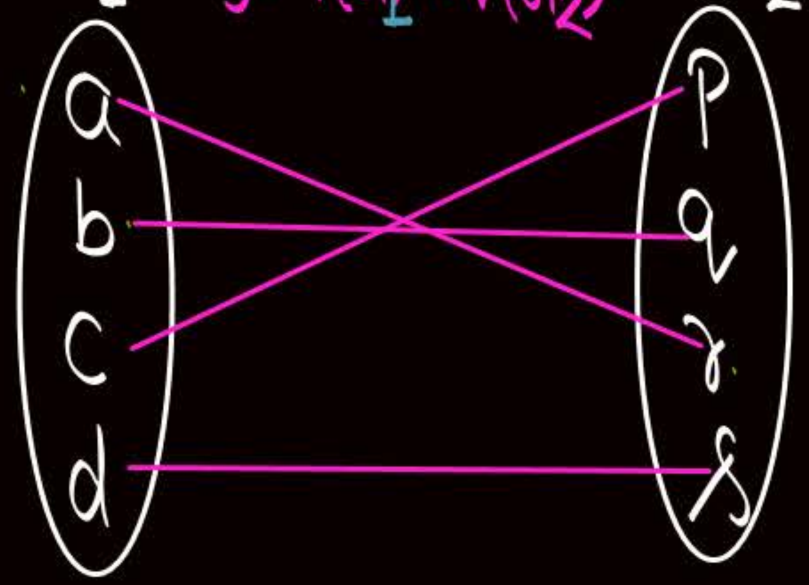


$G_2 (W_4)$

$V(G_1) = \{a, b, c, d\}$

$V(G_2) = \{p, q, r, s\}$

$V(G_1)$ $\quad f: V(G_1) \to V(G_2)$ $\quad V(G_2)$



$f(a) = r$
$f(b) = q$
$f(c) = p$
$f(d) = s$

$$\boxed{\begin{array}{l} \text{(1) Function } f: V(G_1) \to V(G_2) \text{ is bijective} \\ \text{\& (2) 'f' preserves the adjacency of vertices} \end{array}}$$

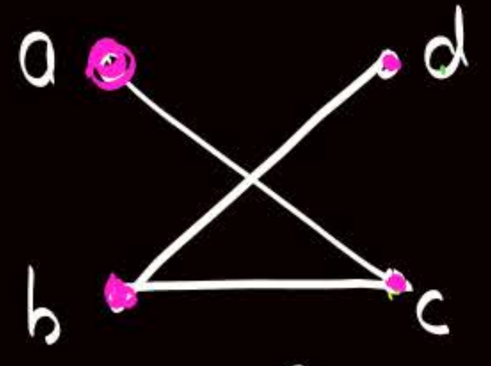
Hence, G_1 & G_2 are isomorphic

G is isomorphic to G' is denoted by

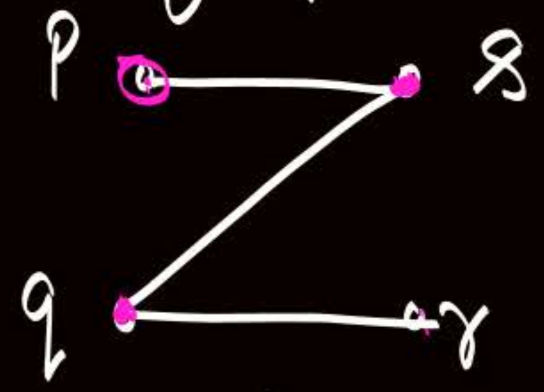
$$G \cong G'$$

"isomorphic"

Q: Consider two graphs G_1 and G_2.
Check whether the graphs are isomorphic or not



G_1



G_2

$|V(G_1)| = |V(G_2)|$

& $|E(G_1)| = |E(G_2)|$

∴ Graphs may or may not be isomorphic

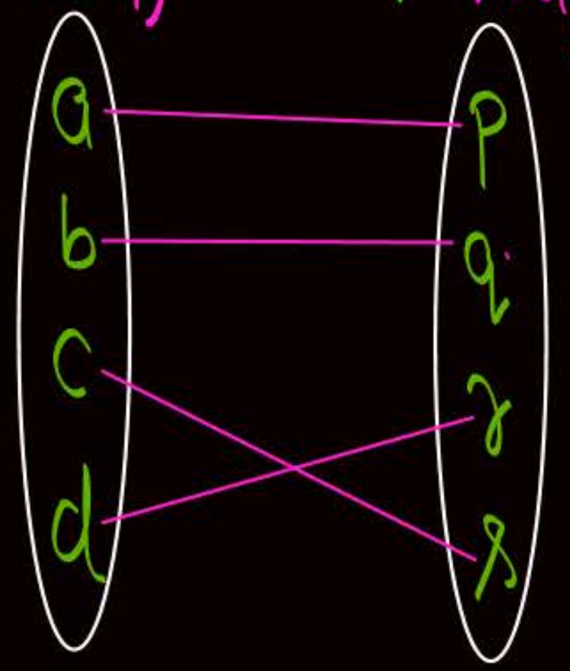
↓

Given graphs G_1 & G_2 are isomorphic

$V(G_1) = \{a, b, c, d\}$

$V(G_2) = \{p, q, r, s\}$

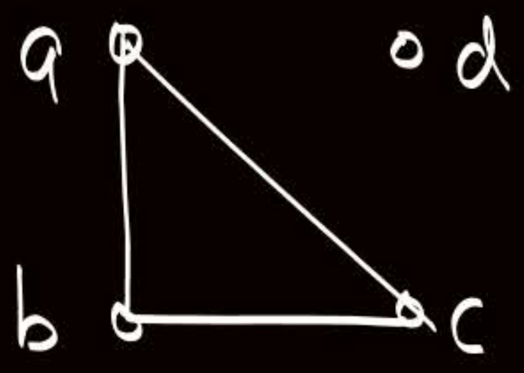
$f : V(G_1) \longrightarrow V(G_2)$



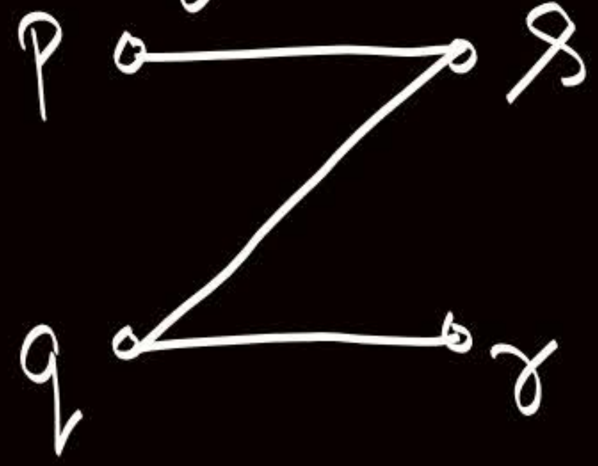
'f' is bijective,
& 'f' preserves adjacency of vertices

∴ G_1 & G_2 are isomorphic

Q: Consider two graphs G_1 and G_2.
Check whether the graphs are isomorphic or not



G_1



G_2

$$\left.\begin{aligned} |V(G_1)| &= |V(G_2)| \\ |E(G_1)| &= |E(G_2)| \end{aligned}\right\}$$

Both Cond^s hold true, but graphs are not isomorphic

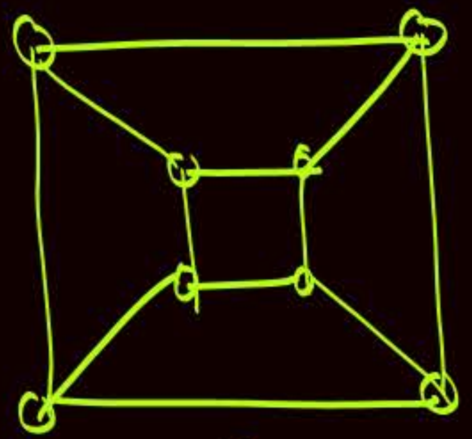
Topic : Graph Isomorphism

If $G \cong G'$, then following conditions must hold true.

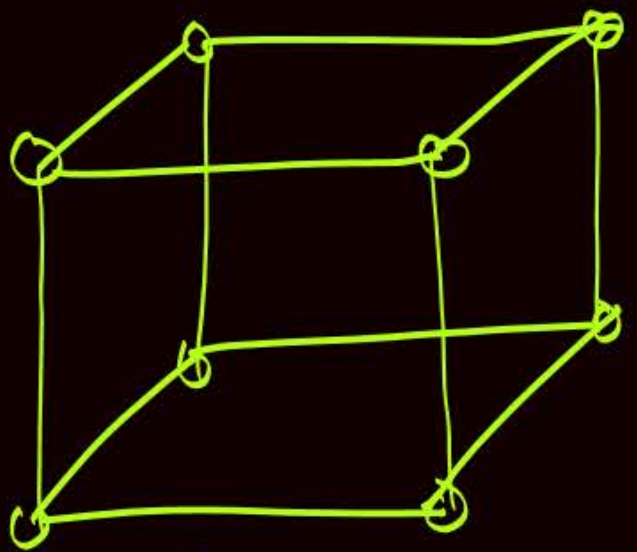
If any of these four conditions is not satisfied, then graphs can not be isomorphic to each other

1. $|V(G)| = |V(G')|$
2. $|E(G)| = |E(G')|$
3. Degree sequence of G = Degree sequence of G'
4. If some 'k' vertices $v_1, v_2, \ldots v_k$ in graph G form a cycle of length 'k', then their corresponding images in G' i.e. $f(v_1), f(v_2), f(v_3), \ldots f(v_k)$ must also form a cycle of length 'k'.

Q: Consider two graphs G_1 & G_2



G_1

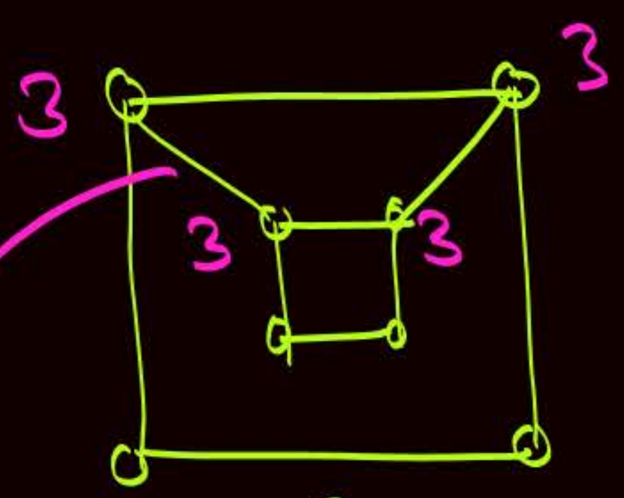


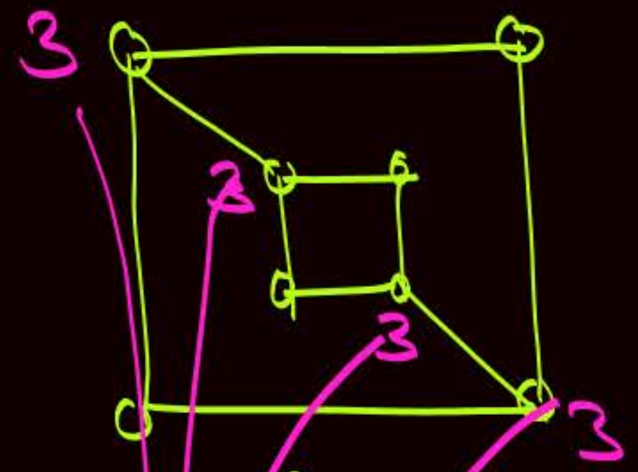
G_2

Check whether G_1 & G_2 are isomorphic or not?

We can observe that $G_1 \cong G_2$

Q: Consider two graphs G_1 & G_2





$|V(G_1)| = |V(G_2)|$

$|E(G_1)| = |E(G_2)|$

Deg. Seq. of G_1 = Deg. Seq. of G_2

$G_1 : \{3, 3, 3, 3, 2, 2, 2, 2\}$ $\quad$ $G_2 = \{3, 3, 3, 3, 2, 2, 2, 2\}$

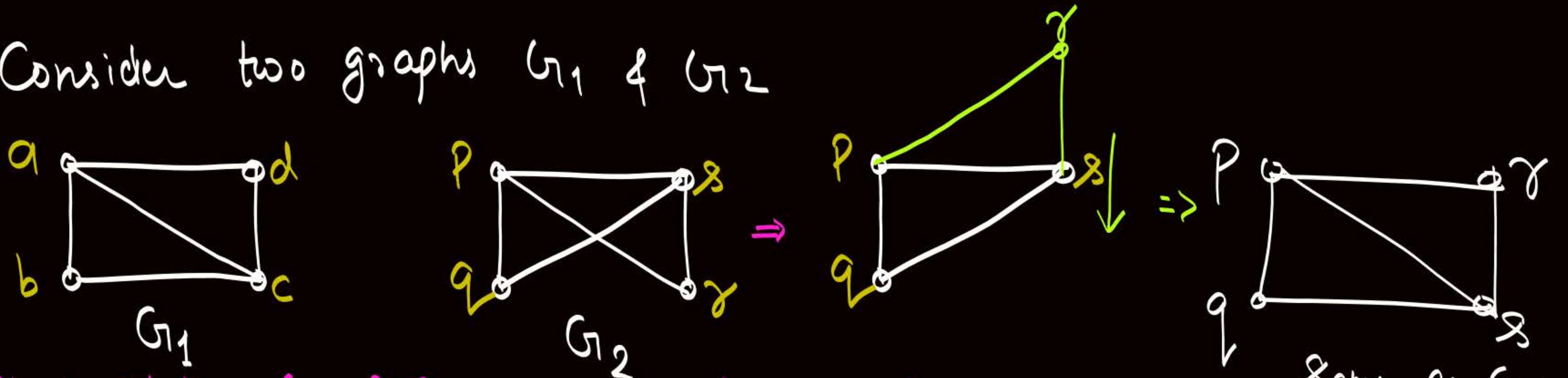
Check whether G_1 & G_2 are isomorphic or not?

Cycle of length '4'

No cycle of length '4'

$\therefore G_1 \ncong G_2$

Q: Consider two graphs G_1 & G_2



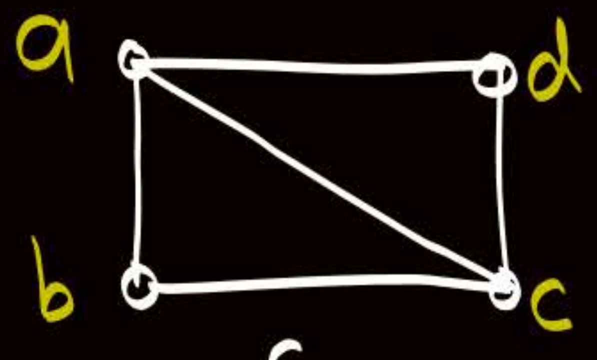
Check whether G_1 & G_2 are isomorphic or not?

same as G_1

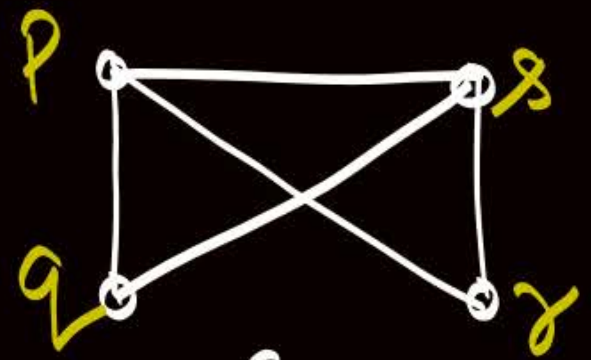
$\therefore G_1 \cong G_2$

Note:- Two simple graphs G_1 and G_2 are isomorphic to each other if and only if their complements are isomorphic to each other

Q:: Consider two graphs G_1 & G_2

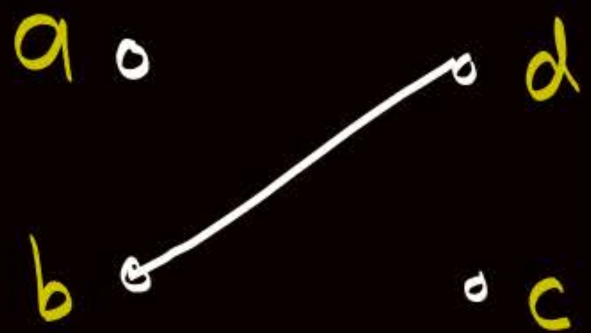


G_1

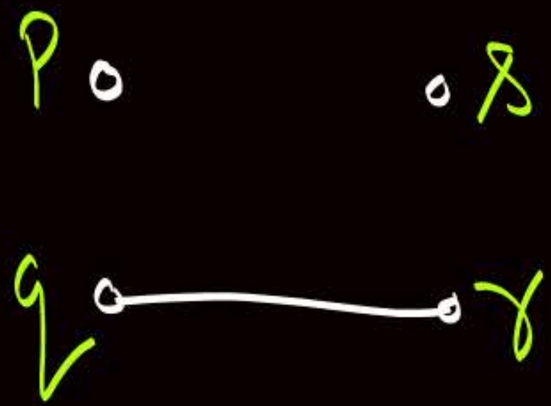


G_2

Check whether G_1 & G_2 are isomorphic or not?



$\overline{G_1}$



$\overline{G_2}$

We can observe

$$\overline{G_1} \cong \overline{G_2}$$

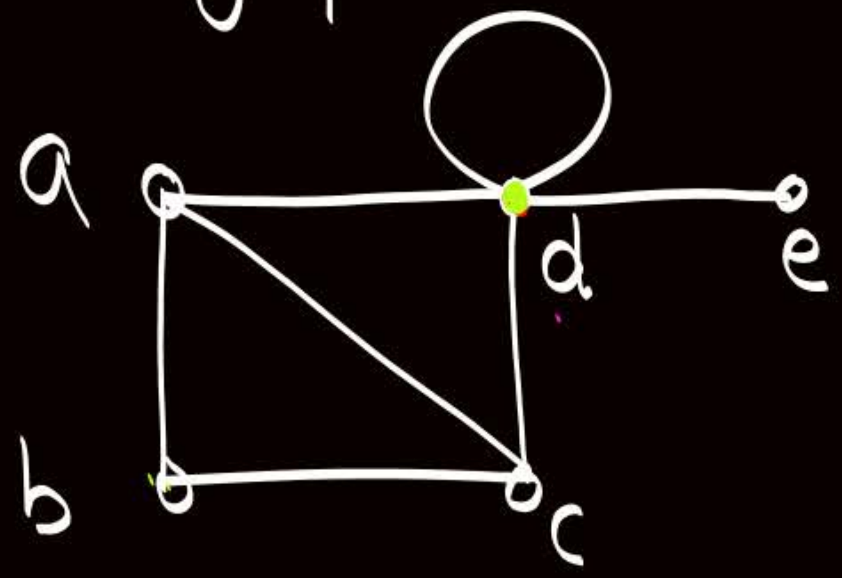
∴ We can say

$$G_1 \cong G_2$$

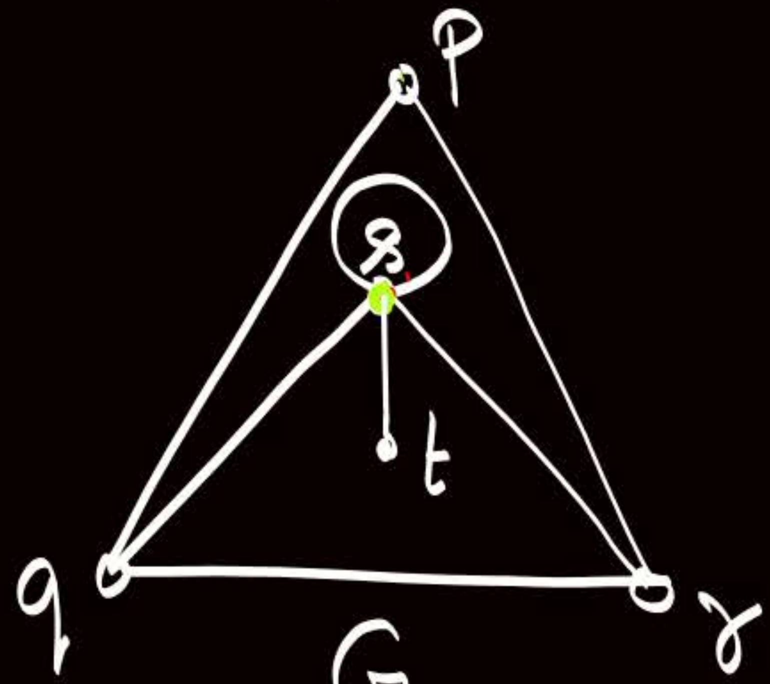
Topic : Graph Isomorphism

* Two graphs G_1 and G_2 are isomorphic if and only if their corresponding subgraphs {i.e the graph obtained by deleting some vertices from graph G_1 and the graph obtained by deleting the the images of those vertices from G_2} are isomorphic to each other

Q: Consider two graphs G_1 and G_2, and check whether the graphs are isomorphic or not?



G_1



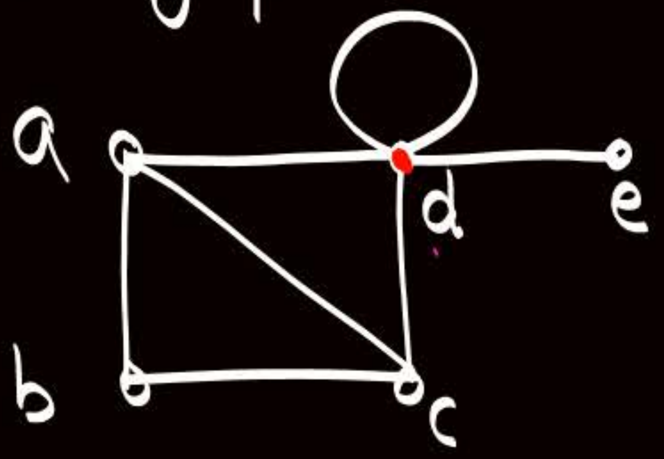
G_2

$|V(G_1)| = |V(G_2)|$

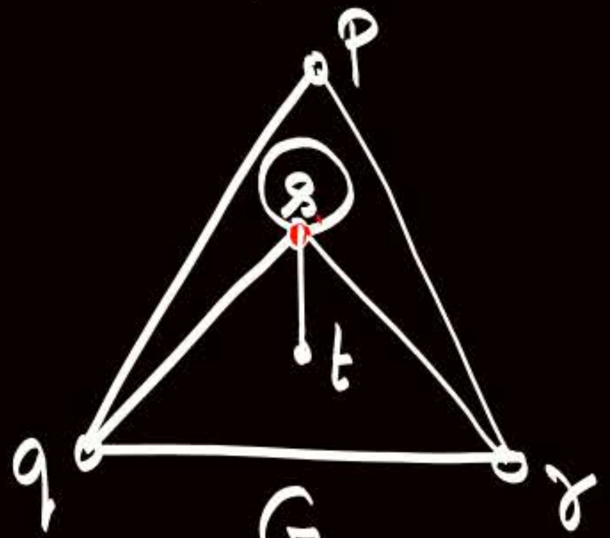
$|E(G_1)| = |E(G_2)|$

Deg. seq of G_1 = Deg seq of G_2

Q: Consider two graphs G_1 and G_2, and check whether the graphs are isomorphic or not?



G_1



G_2

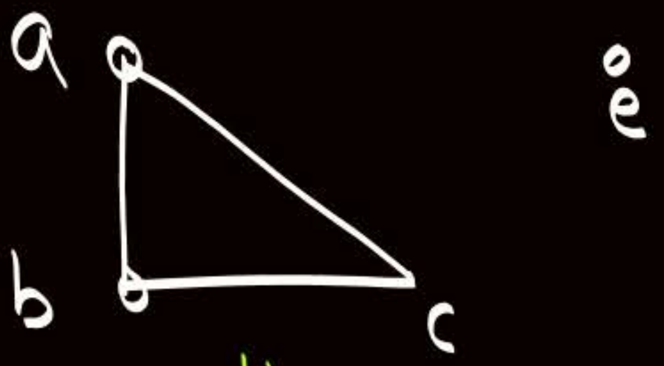
$|V(G_1)| = |V(G_2)|$

$|E(G_1)| = |E(G_2)|$

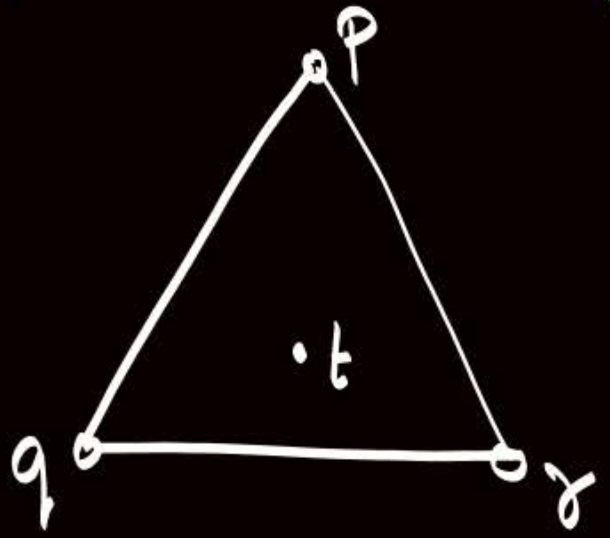
Deg. Seq of G_1 = Deg Seq of G_2

Properties of 'd' in G_1 are same as properties of 's' in G_2 $\therefore$ image of 'd' is 's'

Let H_1 is subgraph of G_1 on deleting 'd', and H_2 is subgraph of G_2 on deleting image of 'd' i.e on deleting s



H_1



H_2

We can observe that $H_1 \cong H_2$

$\therefore G_1 \cong G_2$

Q: How many simple graphs are possible with '3' labeled vertices

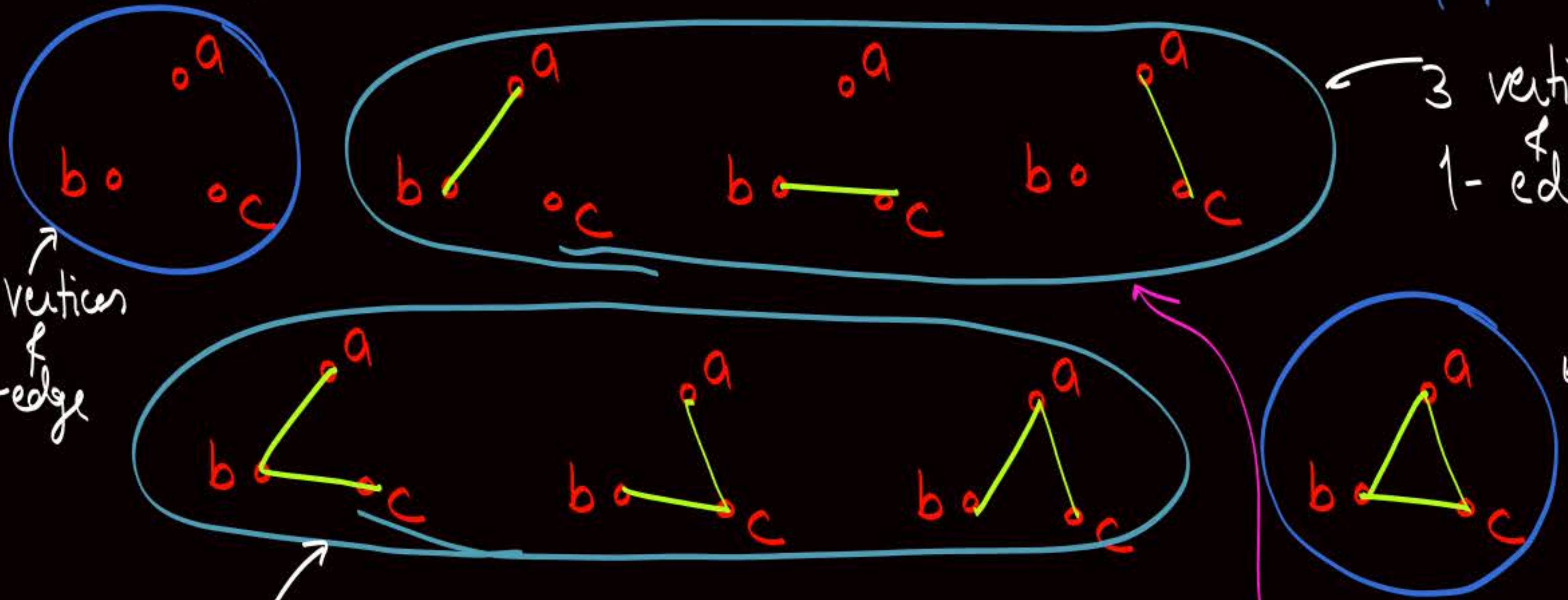
Solun.

$$(2)^{^3C_2} = 2^3 = 8$$

Q:- How many simple "non-isomorphic" graphs are possible with '3' labeled vertices.

Q:- How many simple "non-isomorphic" graphs are possible with '3' labeled vertices.

{ Maximum No. of Edges possible with 3 vertices: ${}^{3}C_{2} = 3$



3 vertices & 0-edge

3 vertices & 1-edge

3-vertices & 2-edges

3 vertices & 3-edges

(All are isomorphic to each other ∴ Counted as only '1' non-isomorphic graphs)

(All are isomorphic to each other ∴ Counted as only '1' non-isomorphic graphs)

Total no. of non-isomorphic graphs with '3' vertices = 4

Q:- How many simple graphs are possible with '4' labeled vertices.

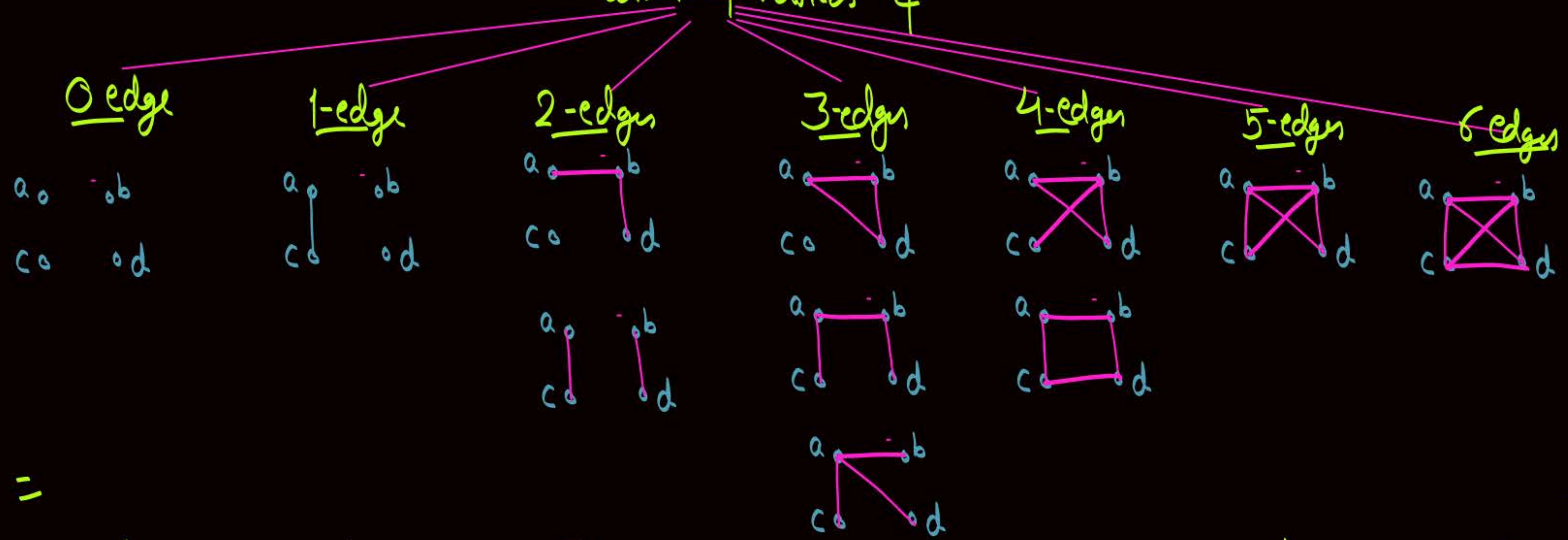
Sol^n $= (2)^{{}^4C_2} = 2^6 = 64$

Q:- How many simple "non-isomorphic" graphs are possible with '4' labeled vertices.

Q:- How many simple "non-isomorphic" graphs are possible with '4' labeled vertices.

Maximum no. of Edges possible in a simple graph with 4 vertices $= {}^4C_2 = 6$

Non-isomorphic graphs possible with 4 vertices



Total no. of non-isomorphic graphs with '4' labeled vertices $= 1 + 1 + 2 + 3 + 2 + 1 + 1 = 11$

Ans

Q:- How many simple "non-isomorphic" graphs are possible with n ($n \geq 4$) labeled vertices and two edges.

Ans = 2

→ '1' when two edges are adjacent

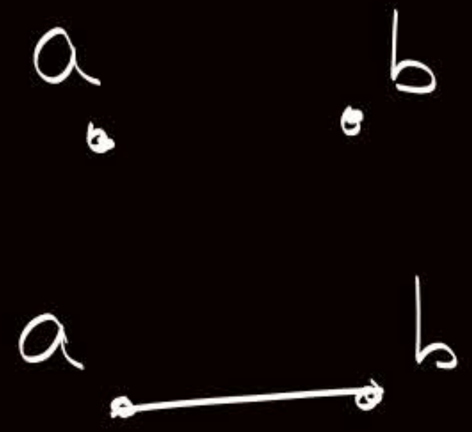
&

→ '1' when the edges are not adjacent

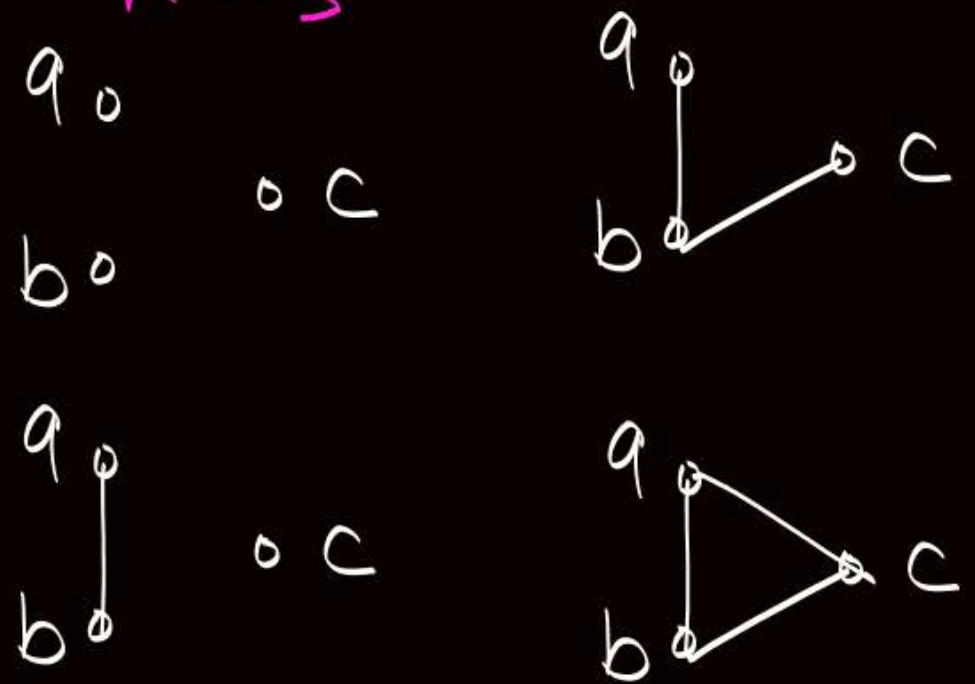
GATE PYQ:- How many different simple graphs are possible with upto '3' vertices.

$n = 1$

$n = 2$



$n = 3$



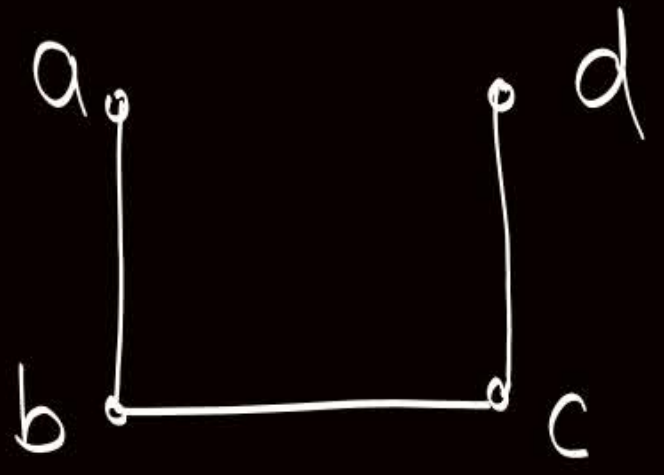
$Ans = 1 + 2 + 4 = 7$

Topic : Self-Complementary Graph

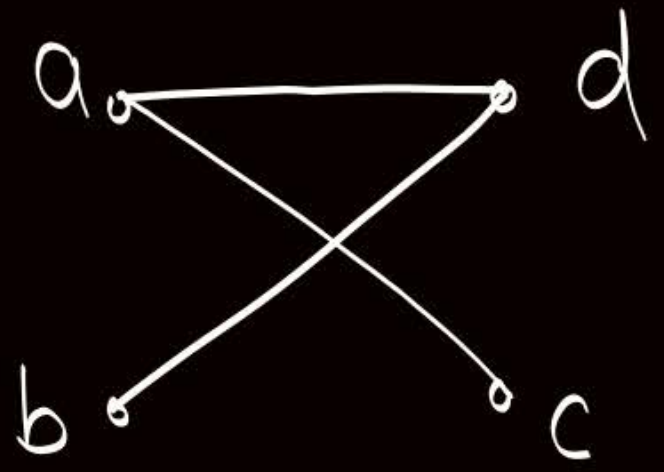
A simple graph G is said to be self-complementary if and only if complement of G is isomorphic to graph G itself.

Graph G is self Complementary
if and only if $G \cong \overline{G}$

eg: Let



G



$\overline{G}$

We can observe $G \cong \overline{G}$

$\therefore$ graph G is a self-complementary graph

Note

In a $\underbrace{\text{self complementary graph } G}_{\text{i.e. when } G \cong \overline{G}}$ with 'n' vertices

① $|E(G)| = |E(\overline{G})|$ because of isomorphism —— eqn ①

② G & $\overline{G}$ are complement of each other

$\therefore \quad |E(G)| + |E(\overline{G})| = \dfrac{n(n-1)}{2}$ —— eqn ②

By eqn ① & eqn ②

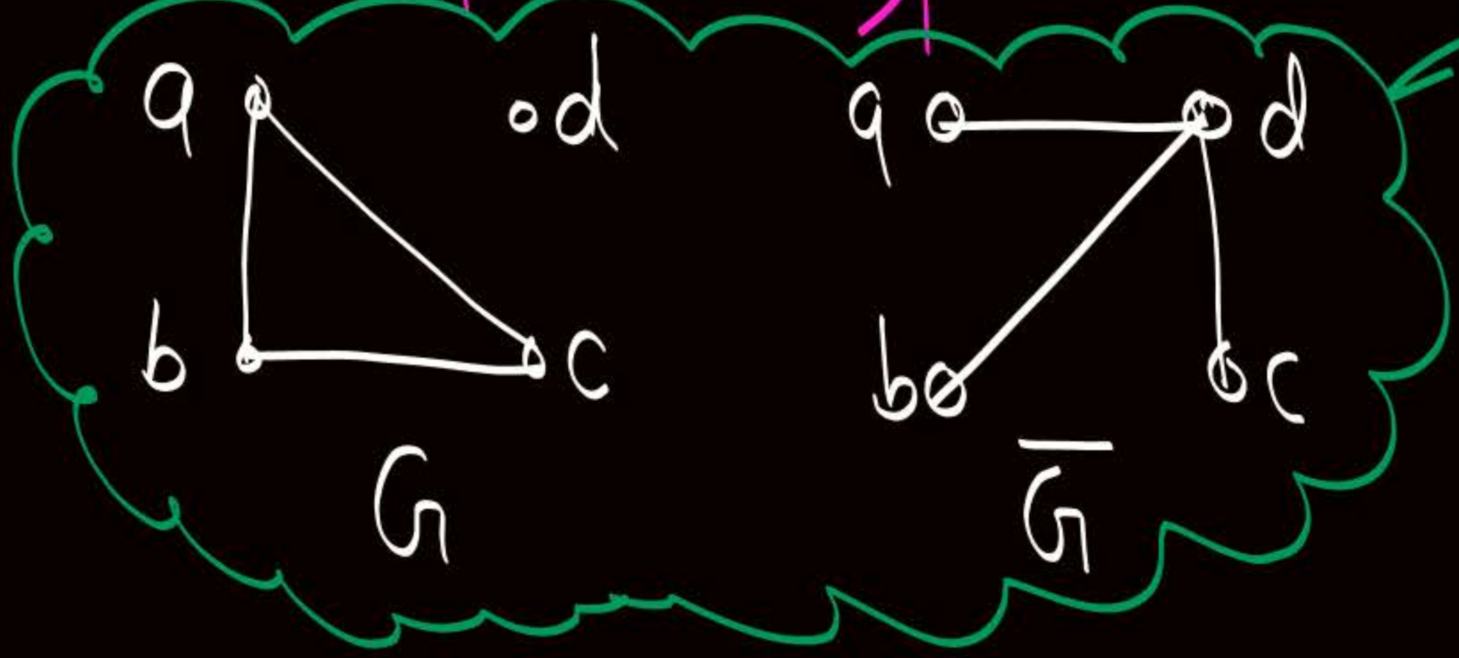
$$|E(G)| + |E(G)| = \frac{n(n-1)}{2}$$

$$\boxed{|E(G)| = \frac{n(n-1)}{4}}$$

Note: If G is a self complementary graph with 'n' vertices then $|E(G)| = \frac{n(n-1)}{4}$, but converse of the statement need not be true

eg let $n = 4$

$\therefore \frac{n(n-1)}{4} = \frac{\cancel{4}\cdot(4-1)}{\cancel{4}} = 3$



$|E| = \frac{n(n-1)}{4}$, but graph is not a self-complementary graph

Note: In a self Complementary graph 'G'

$|E(G)| = \frac{n(n-1)}{4}$, and number of edges can never be in fraction.

$\therefore$ If graph G is a self Complementary graph then $\frac{n(n-1)}{4}$ must be an integer

$\therefore$ either 'n' must be multiple of '4'

or

'n-1' must be multiple of 4

i.e. $n = 4k$ or $n = 4k+1$

where k is any positive integer

Note:

$\boxed{\begin{array}{l}\text{In a self-Complementary graph } G, \\ \quad |V(G)| \bmod 4 = 0 \text{ or } 1\end{array}}$

Q:- Let 'C_n' represent a cycle graph with 'n' vertices. If 'C_n' is a self complementary graph, then find the value of 'n'.

- C_n is a cycle graph $\therefore$ $|E(C_n)| = n$ —— (1)
- $C_n \cong \overline{C_n}$ $\therefore$ $|E(C_n)| = \dfrac{n(n-1)}{4}$ —— (2)

By (1) & (2)

$$n = \frac{n(n-1)}{4} \Rightarrow \boxed{n = 5}$$

Q:- Let 'Cn' represent a cycle graph with 'n' vertices, If 'Cn' is a self complementary graph, then find the value of 'n'.

* C_n is a cycle graph
  and we know $|E(C_n)| = n$ — eq^n ①

* C_n is a self-complementary graph
  $\therefore |E(C_n)| = \frac{n(n-1)}{4}$ — eq^n ②

By eq^n ① & ②

$$n = \frac{n(n-1)}{4}$$

$$\boxed{n = 5}$$

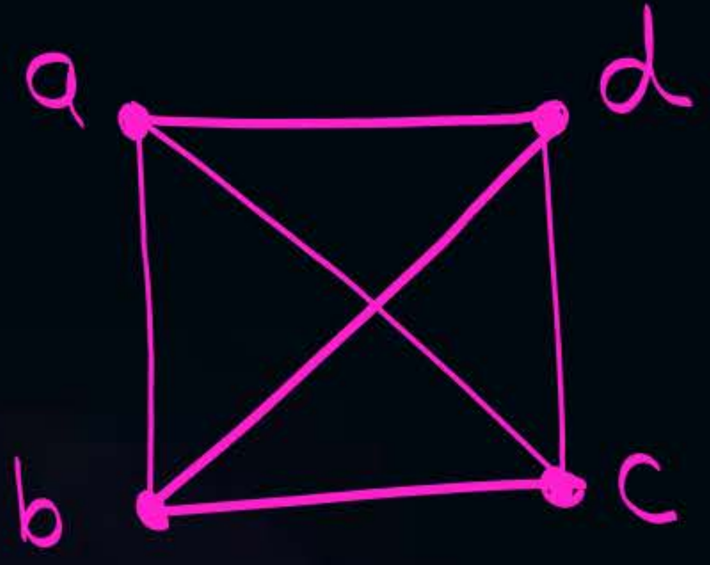
Planar Graph

Topic : Planar graph

A graph G= (V, E) is said to be planar if it can be drawn in the plane so that no two edges of G intersect each other at a non-vertex point.

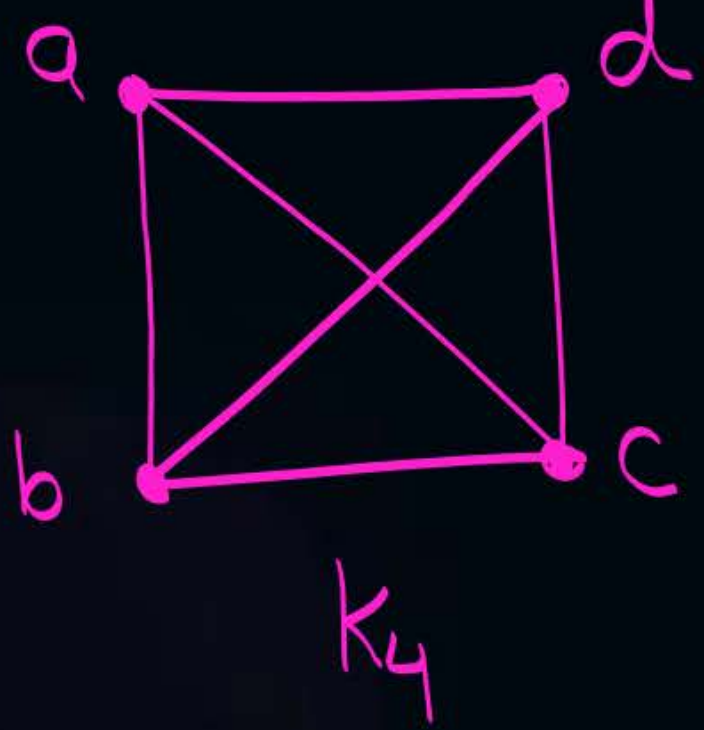
Such a drawing of a planar graph is called a planar embedding of the graph.

Topic : Planar graph

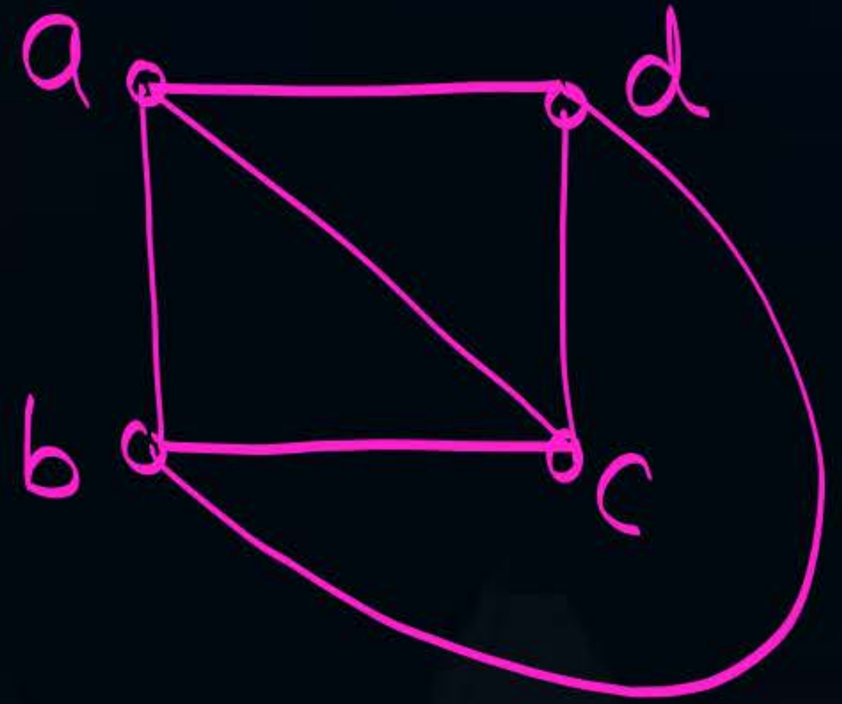


K_4

Topic : Planar graph



$\Longrightarrow$

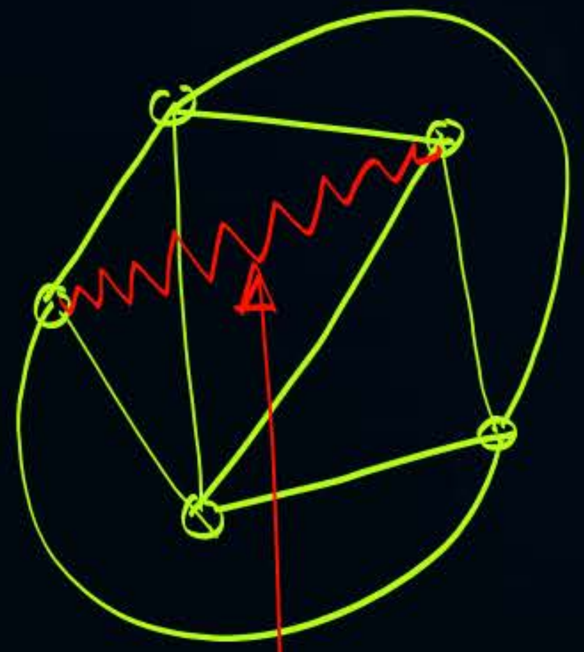


We can draw K_4 on a plane s.t. no two edges cross each other at a non-vertex point.

$\therefore$ K_4 is planar

Topic : Planar graph

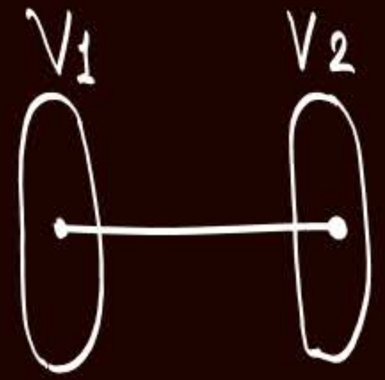
K_1 K_2 K_3 K_4



Complete graph K_n is planar if and only if $n \leq 4$

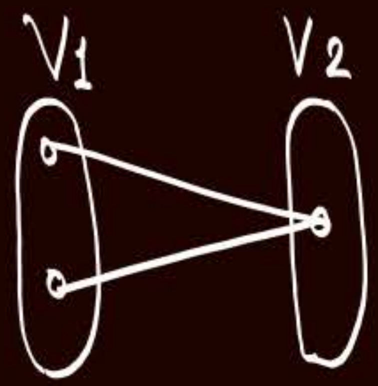
K_5 is non-planar

- K_5 is a non-planar graph with 5 vertices & 10 edges



$K_{1,1}$

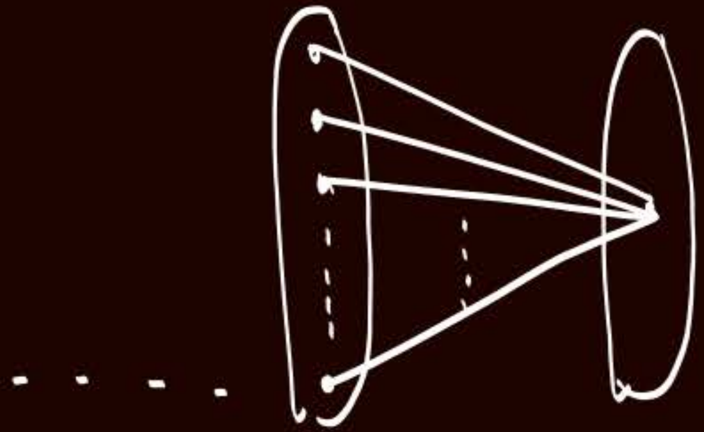
is planar



$K_{2,1}$

is planar

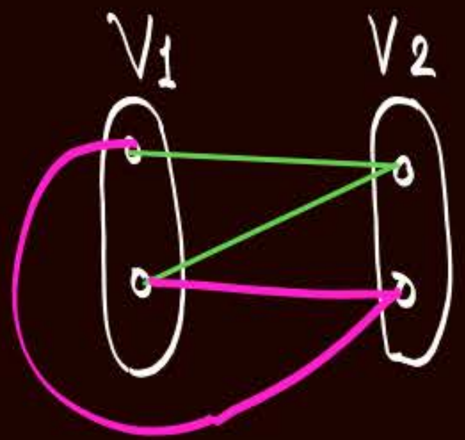
∴ $K_{1,2}$ is also planar



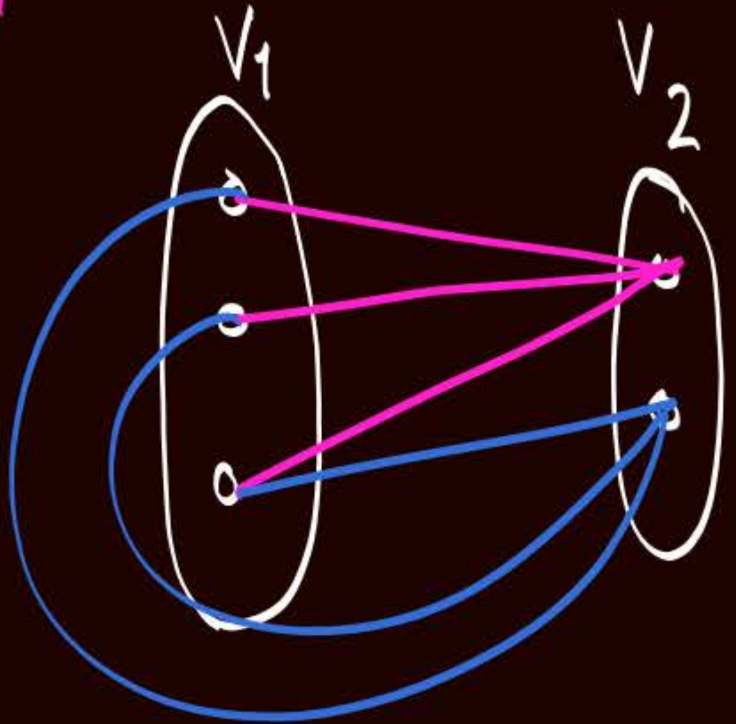
$K_{n,1}$

is planar

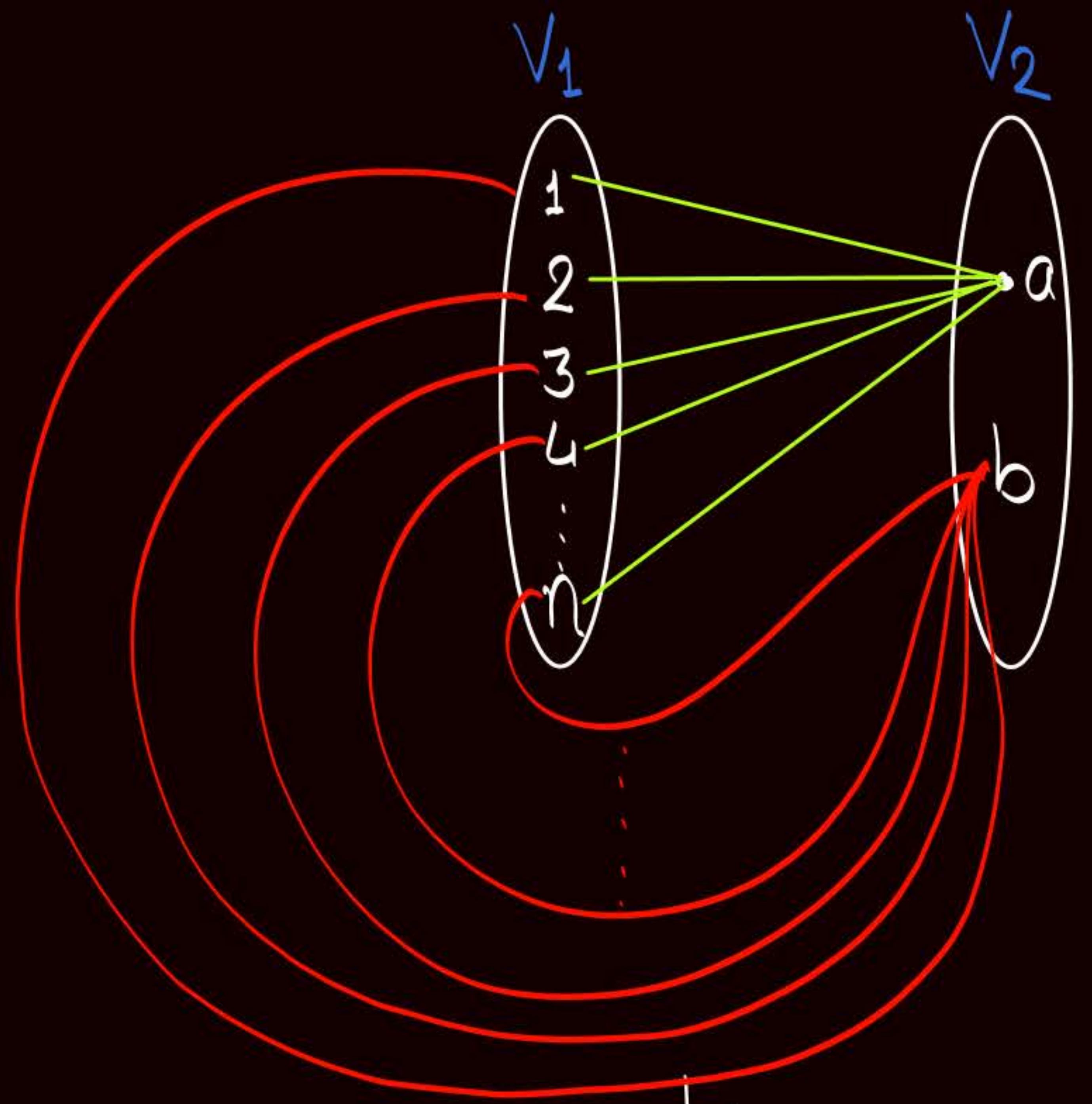
∴ $K_{1,n}$ is also planar



$K_{2,2}$ is planar



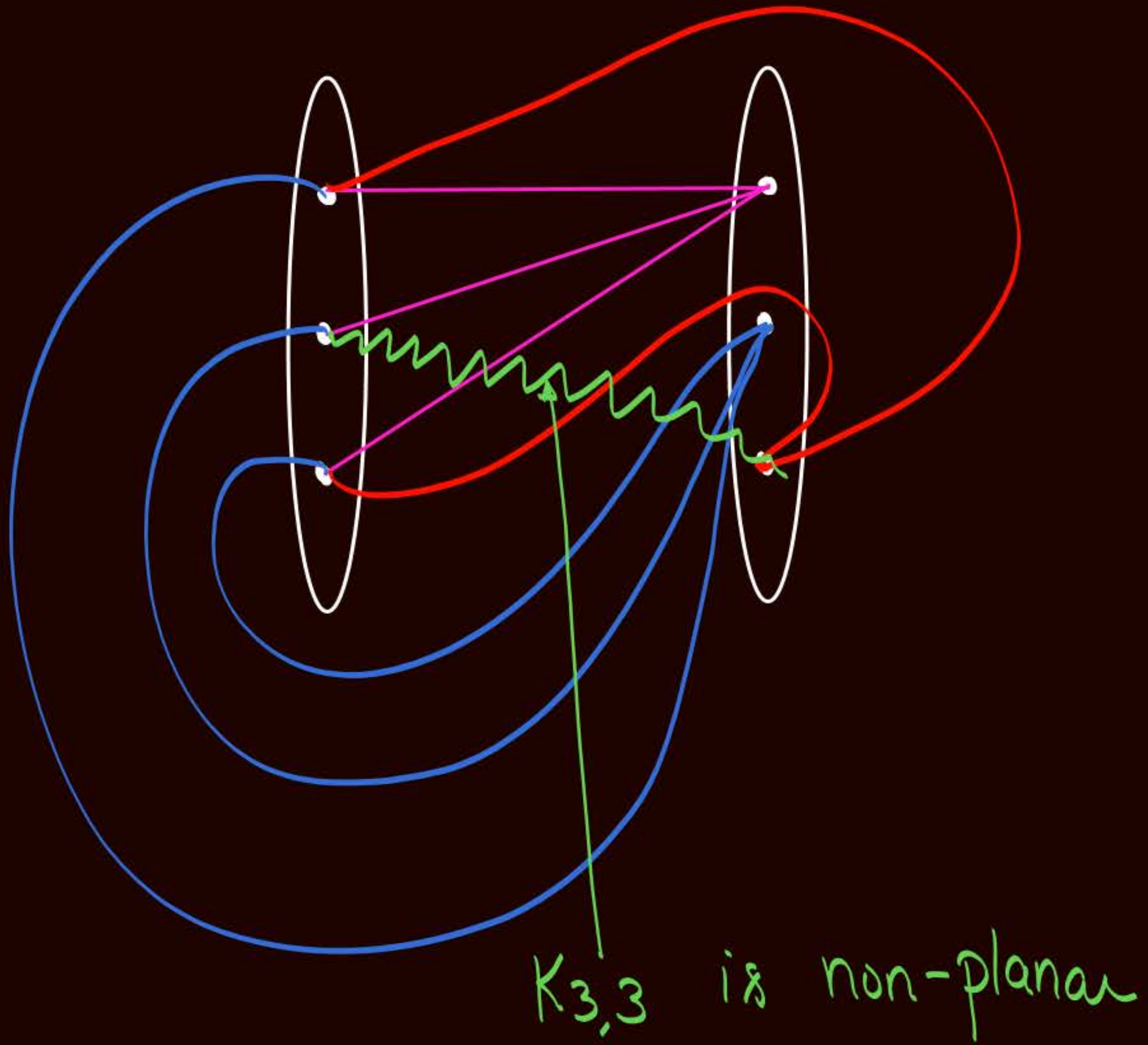
$K_{3,2}$ is planar



$K_{n,2}$ is planar

$\therefore K_{2,n}$ is also planar

* Check whether $K_{3,3}$ is a planar graph or not?



$K_{3,3}$ is a non-planar graph with '6' vertices but only '9' edges

Note: Complete bi-partite graph $K_{m,n}$ is planar if and only if $m \le 2$ or $n \le 2$

Note :-

- (1) K_5 is the non-planar graph with minimum number of vertices.

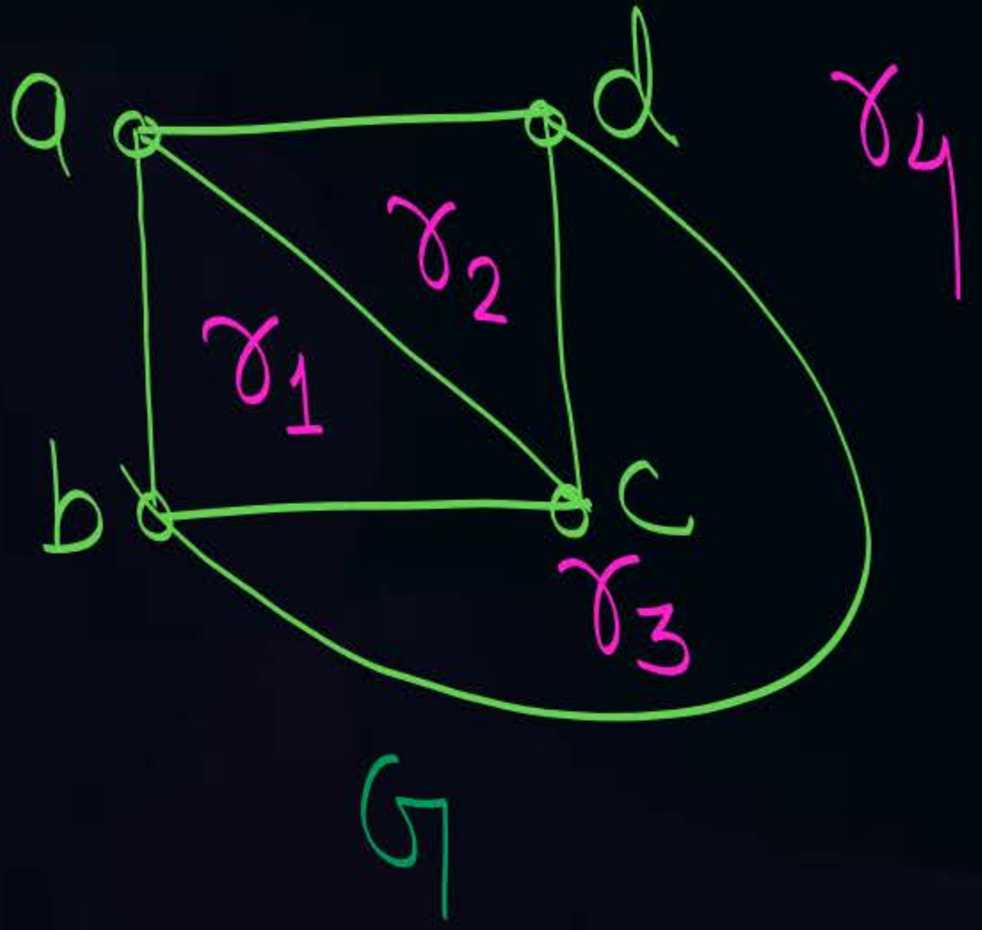
  it has only 5 vertices but 10 edges

- (2) $K_{3,3}$ is a non-planar graph with minimum number of edges.

  it has only 9 edges, but 6 vertices

Topic : Region / Face

* Every planar graph divides a plane into connected areas called as regions of the plane or faces of the planar graph.



* r_1, r_2 and r_3 are bouded regions (interior regions) of planar graph G

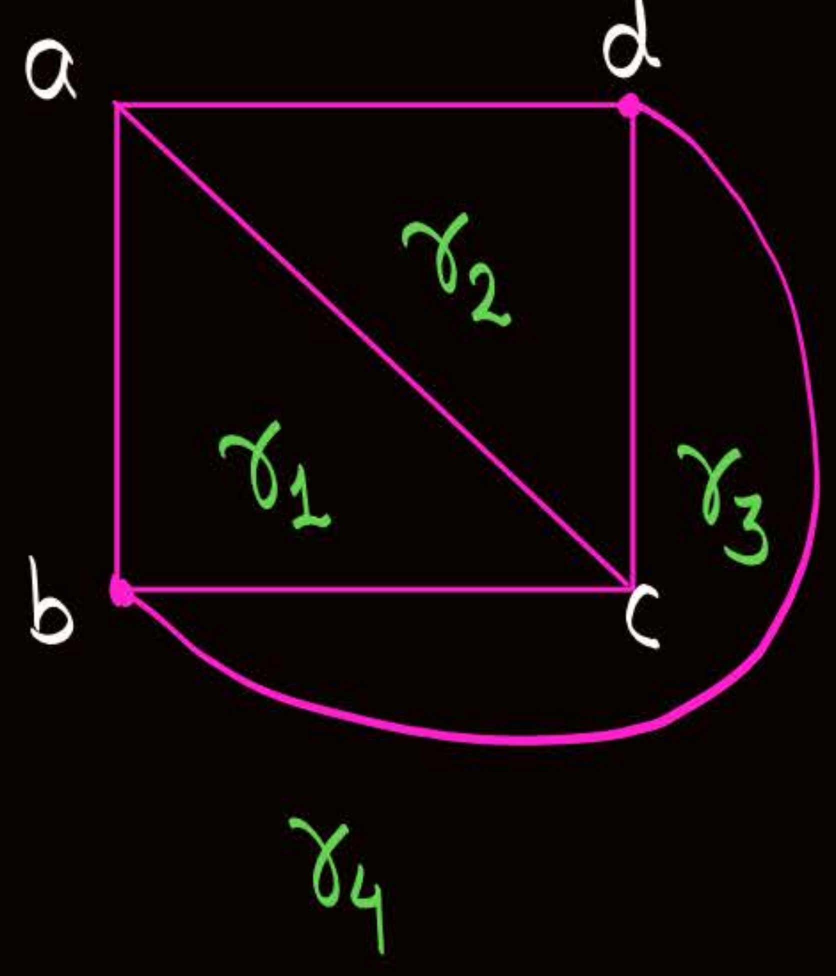
Whereas r_4 is an unbounded (Exterior) region of planar graph G.

Topic : Region / Face

- In any planar graph there will be exactly '1' Unbounded (Exterior) region, all other regions of the Planar graph will be bounded (interior) regions.

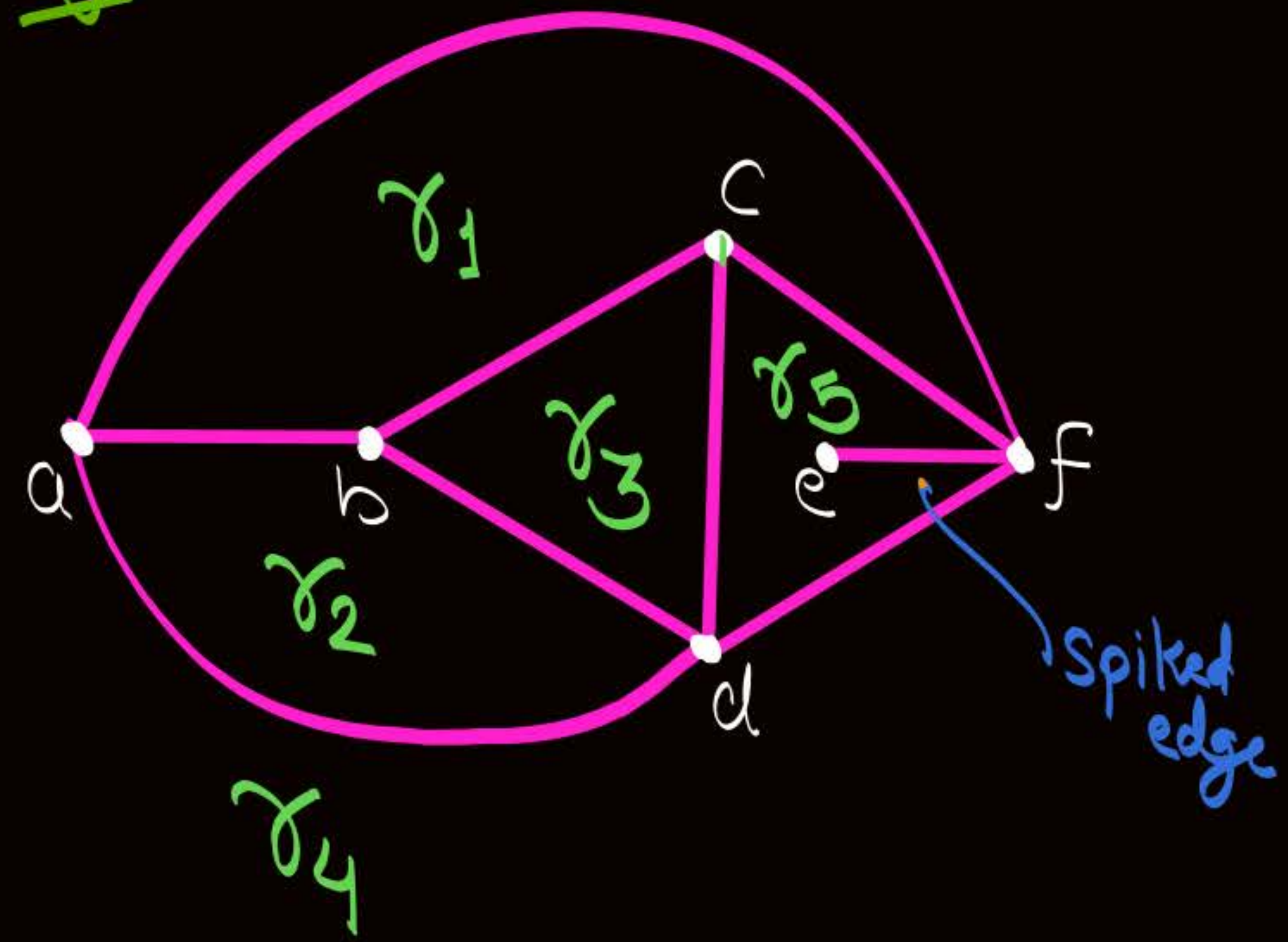
- Number of bounded regions (faces) in a planar graph = Total no. of regions in that Planar graph − 1 eg 1:

G_1



eg 2:-

G_2



Topic : Degree of a region/face

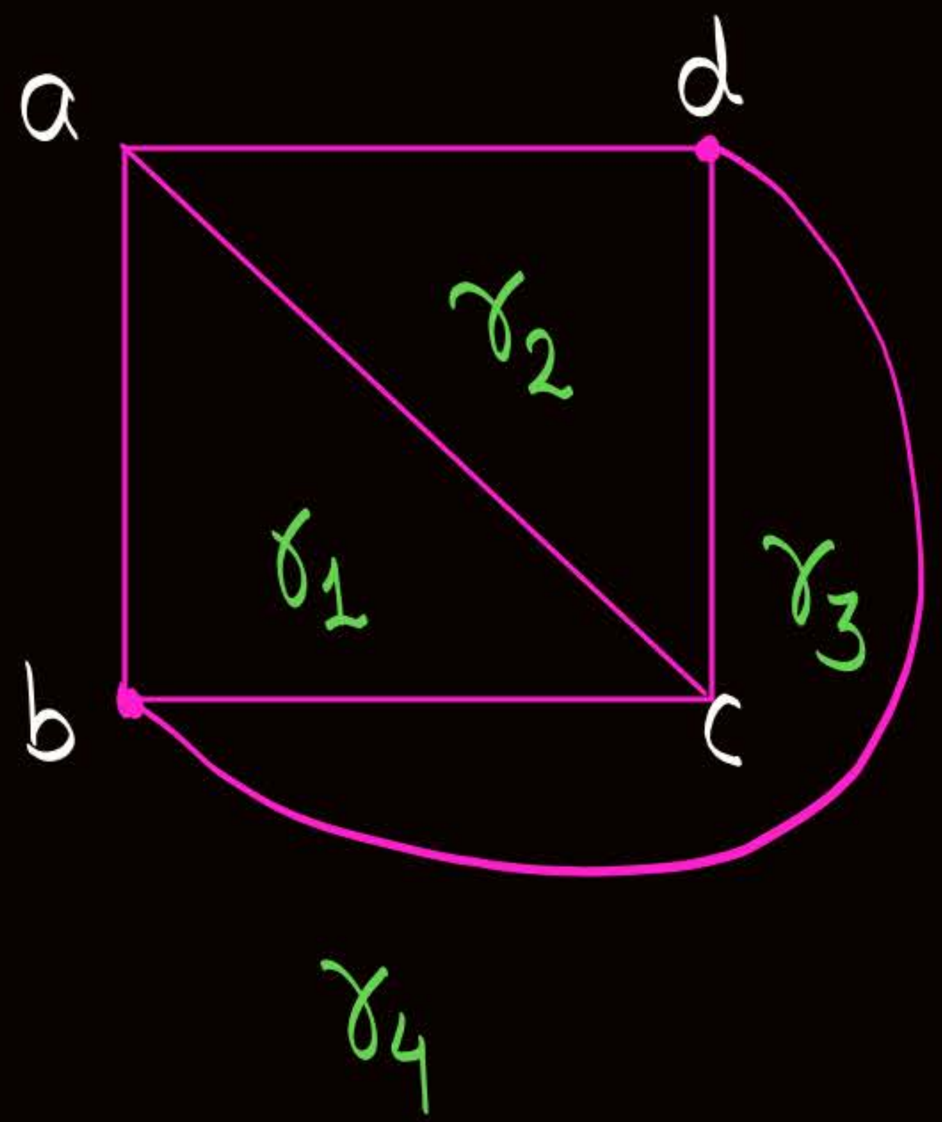
* Degree of a region "r" can be denoted by deg(r), and it is defined as.

$$\deg(r) = \text{No. of edges Enclosing the region 'r'}$$

(or)

No. of Edges Exposed to region 'r'

eg:-



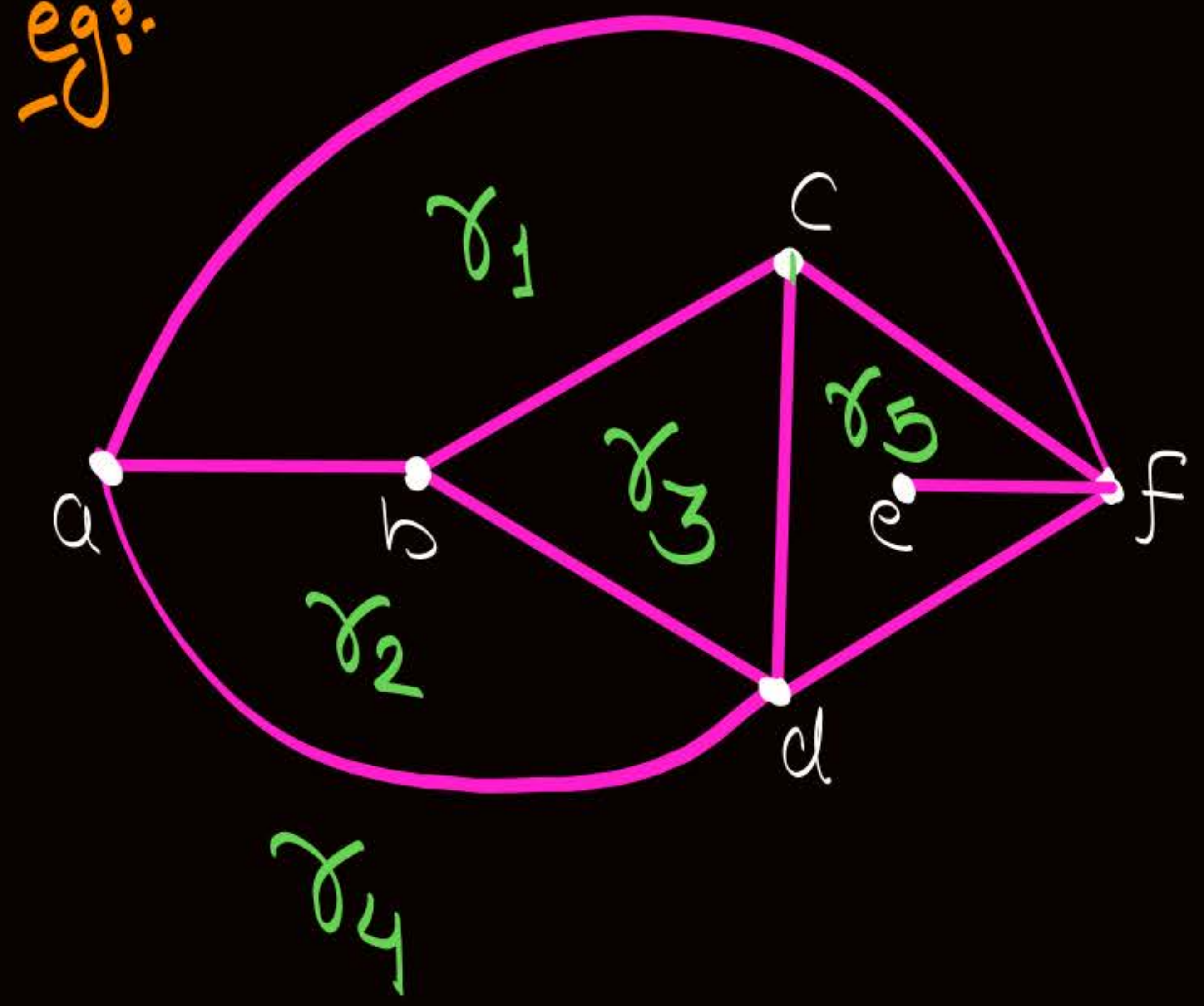
$\deg(r_1) = 3$

$\deg(r_2) = 3$

$\deg(r_3) = 3$

$\deg(r_4) = 3$

eg:



Note: Spiked edge will be Counted as two edges to obtain the degree of Corresponding region

$\deg(r_1) = 4$

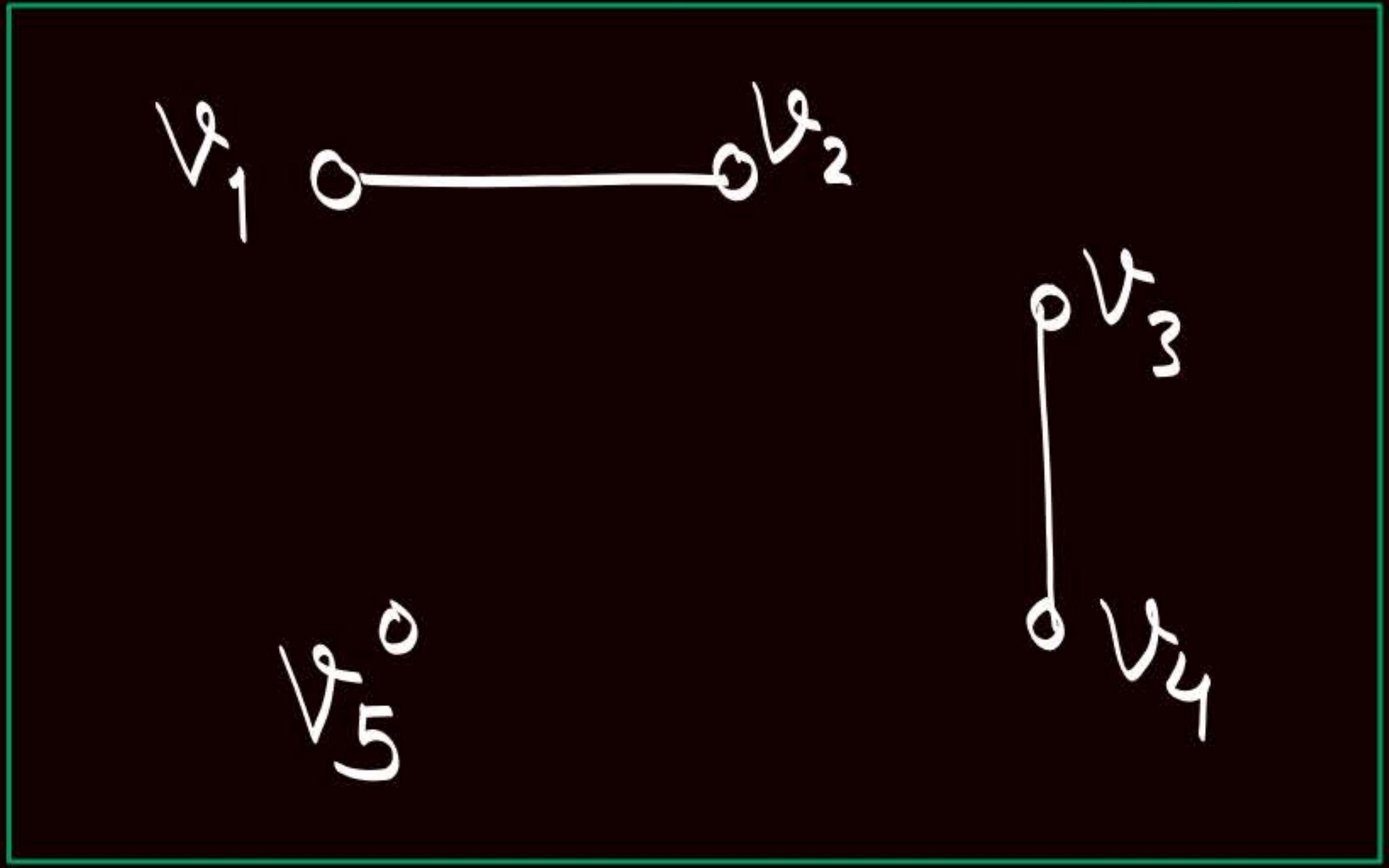
$\deg(r_2) = 3$

$\deg(r_3) = 3$

$\deg(r_4) = 3$

$\deg(r_5) = 3 + \underset{\text{w.r.t. Spiked Edge}}{2} = 5$

Consider the following graph G



it is a planar graph with only '1' region, let's call it region 'r', then

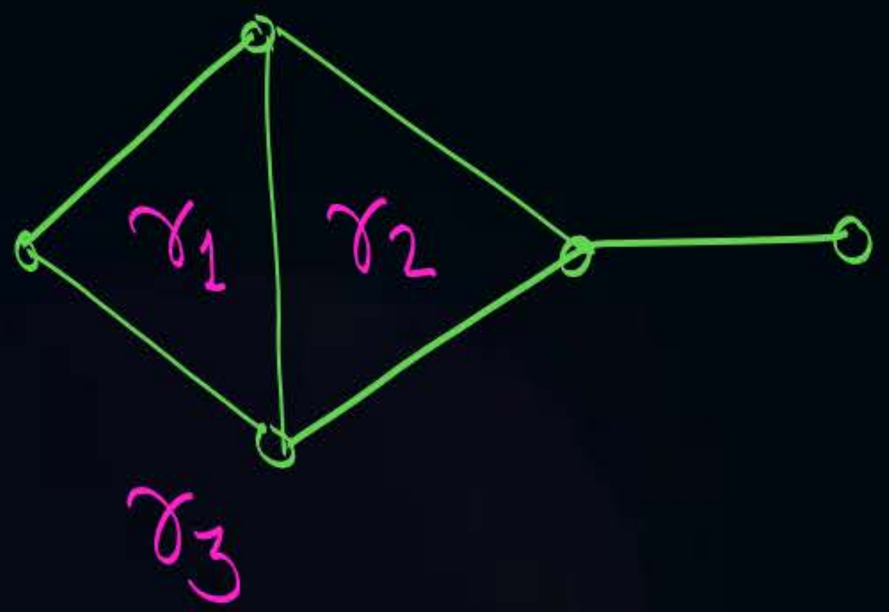
$$\deg(r) = 2 + 2 = 4$$

No two edges cross each other at a non-vertex point,

$\therefore$ graph G is a planar graph

Topic : Sum of degree theorem

W.r.t. degree of regions



$\text{deg}(r_1) + \text{deg}(r_2) + \text{deg}(r_3)$

$3 + 3 + 6 = 12$

$= 2 \times 6 = 2 \times |E|$

* Let $|R|$ denotes the number of regions in the planar graph $G = (V, E)$, then

$$\sum_{i=1}^{|R|} \text{deg}(r_i) = 2|E|$$

Some authors denotes the no. of regions by $|R|$ and some author denote the no. of faces by 'f' and $|R| = f$

Topic : Sum of degree theorem

* Corollary 1 :- No. of regions of odd degree are always even

* Corollary 2 :- In a planar graph G, if degree of each region is exactly 'K' then

$$\boxed{K \cdot |R| = 2|E|}$$

Corollary 3: In a planar graph G if degree of each region is at least 'K' {i.e. $\geq K$} then

$$\boxed{K|R| \leq 2|E|}$$

Topic : Sum of degree theorem

Corollary 4: In a planar graph G if degree of each region is at most 'K' {i.e $\leq K$} then

$$\boxed{K|R| \geq 2|E|}$$

2 mins Summary

Topic Graph isomorphism

Topic Self-complementary graph

Topic Planar graphs

Slide

THANK - YOU

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